

Genetic and ecological assessment of population rehabilitation: walleye in Lake Superior

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Abstract. Sustainable management of exploited species is an ongoing challenge, particularly where populations have collapsed or been depleted by overharvest and habitat alteration. The walleye (*Sander vitreus*) population in Lake Superior's Black Bay historically supported more than 90% of the commercial walleye harvest from the entire lake, but collapsed in 1968 and has still not recovered despite long-term closure of the fishery. In an effort to rehabilitate this population, hatchery-origin walleye from exogenous sources were released into Black Bay between 2003 and 2005. We used individual-based analysis of genetic data collected between 2007 and 2010 to examine the contributions of different wild sources and hatchery stocking events to the contemporary walleye population in Black Bay. We found that 75% of the walleye in Black Bay originated from above- and below-barrier native populations in the Black Sturgeon River. The hatchery stocking events differed considerably in their effectiveness: the 2003 release of fry had no measurable contribution, whereas the 2004 and 2005 releases of fingerlings contributed 71% and 45% of the fish in their respective age classes. Hatchery and wild fish were similar in size, but hatchery fish rarely utilized the river habitat where Black Bay walleye historically spawned, and there was little genetic evidence of interbreeding or natural recruitment of stocked fish. Overall, our results suggest that restoring habitat connectivity to facilitate wild recruitment has greater potential than further exogenous stocking to contribute to the recovery of walleye in this system.

Key words: Black Bay, Lake Superior; conservation; dams; genetic analysis; hatchery stocking; population assignment; *Sander vitreus*; walleye population recovery.

INTRODUCTION

Successful rehabilitation of degraded or depleted populations requires a sound biological understanding of the threatened system, as well as the factors responsible for its reduced state. For systems with complex histories of anthropogenic stressors in particular, knowledge about current and historical states can identify contemporary barriers to restoration, and thus prioritize efforts among mitigation strategies that might include reducing exploitation (Hutchings 2000, Worm et al. 2009), restoring habitat (Roni et al. 2008, Kocovsky et al. 2009), or controlling invasive species (Lawrie 1970, Strayer 1999). Genetic information can be used to understand the distribution of biodiversity within a taxon, and thus to identify unique populations that have high conservation value (Moritz 1994). When reintroduction programs are necessary, genetic and ecological similarity to extirpated populations can be used to identify the most suitable source populations for reintroduction (Meffe 1995).

Walleye (*Sander vitreus*) were once abundant in the Black Bay area of Lake Superior, but collapsed in response to multiple anthropogenic stressors. At its peak the commercial fishery in Black Bay harvested more than 125 Mg (metric tons) of walleye annually, which accounted for >90% of all walleye harvested in Lake Superior (Furlong et al. 2006). However, the Black Bay walleye population collapsed in 1968, and despite closure of the commercial fishery in 1971, walleye abundance in Black Bay remains extremely low (Addison and Bobrowicz 2009, Bobrowicz 2010). Historical overexploitation and habitat loss have both been recognized as important contributing factors to this collapse (Colby and Nepszy 1981, MacCallum and Selgeby 1987, Furlong et al. 2006). In its final years the commercial fishery harvested increasingly younger fish, and for several years prior to the collapse catches were more than double the sustainable yield predicted from habitat availability (Colby and Nepszy 1981, Bobrowicz 2010). The loss of spawning habitat has also been implicated as an important contributor to this collapse (MacCallum and Selgeby 1987, Furlong et al. 2006). Anecdotal reports indicate that walleye may spawn on shoals in Black Bay, but suitable spawning substrate is largely absent in Black Bay (Biberhofer and Prokopec

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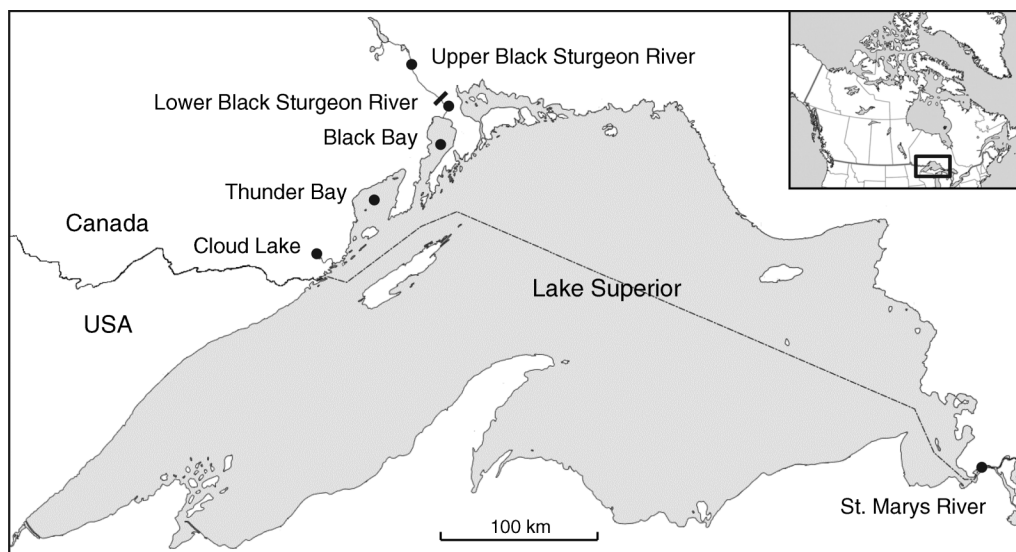


FIG. 1. Map of Lake Superior showing the locations of the walleye (*Sander vitreus*) populations included in the genetic analyses. The location of the Black Sturgeon Dam is marked by a line dividing the populations in the upper and lower Black Sturgeon River.

2007), and radio telemetry during the spring spawning migration observed walleye gathering to spawn only in the Black Sturgeon River (Furlong et al. 2006). In 1960 a logging dam was completed 17 km from the mouth of the Black Sturgeon River (Fig. 1; see Plate 1), which isolated Black Bay walleye from 80% of the spawning habitat historically available to these fish (Furlong et al. 2006, Bobrowicz 2010). It is not certain if walleye in Black Bay are currently limited by access to spawning habitat, but walleye are known to congregate below the Black Sturgeon dam during the spring spawning migration, which is consistent with these fish attempting to utilize upstream habitat (Furlong et al. 2006, Bobrowicz 2010). Moreover, walleye above and below the dam are genetically similar to each other and to the historical walleye population in Black Bay, indicating that these populations were highly connected prior to their collapse (Wilson et al. 2007). Finally, the abundance of rainbow smelt (*Osmerus mordax*) in Lake Superior increased considerably in the 1950s, and rainbow smelt predation has previously been linked to declines in walleye recruitment (Schneider and Leach 1977, Mercado-Silva et al. 2007). The introduction of rainbow smelt into Lake Superior has thus been implicated as another potential contributor to the collapse of Black Bay walleye, although its importance is less established than that of overexploitation and habitat fragmentation.

Given the value of the Black Bay walleye fishery, considerable effort has been expended on its restoration. Because the closure of the Black Bay walleye fishery in 1971 failed to initiate a recovery, subsequent restoration programs have largely focused on exogenous stocking. Two small-scale catch-and-transfer programs were

initially attempted: 1032 adult walleye from the nearby Current River (Thunder Bay) and Pigeon River were transferred to Black Bay in 1972, and 768 adult walleye from local inland lakes were transferred to Black Bay between 1998 and 2000 (Furlong et al. 2006). However, a 2002 survey using a standardized fall walleye index netting (FWIN) protocol produced a catch rate of only 0.16 walleye per net in Black Bay, compared to an average of 10.7 walleye per net for other lakes in the area (Furlong et al. 2006, Addison and Bobrowicz 2009). Thus, these stocking programs were not associated with a measurable improvement in the abundance of walleye in Black Bay. Consequently, a much larger hatchery stocking program was initiated in Black Bay, which led to the release of 1 000 000 fry from Cloud Lake in 2003 (themselves descended from a transfer of adult walleye from the Current River into Cloud Lake in the 1970s) and the release of 260 000 fingerlings from the St. Marys River between 2004 and 2005 (Furlong et al. 2006). In 2008 the abundance of walleye in Black Bay had increased 10-fold to 1.65 walleye per net, with hatchery fish representing ~30% of the population (Addison and Bobrowicz 2009, Bobrowicz 2010). Thus, hatchery stocking appears to be contributing to an increase in the abundance of walleye in Black Bay.

There are, however, a number of uncertainties about the effects of hatchery stocking in Black Bay. First, previous analyses provided only a coarse measure of the abundance of hatchery walleye in Black Bay because they did not isolate the contributions of individual hatchery stocking events or estimate recruitment of hatchery fish to the next generation (Bobrowicz 2010). Second, previous sampling was conducted in a single year and a single habitat (Bobrowicz 2010), so it was not

possible to measure changes in the abundance of hatchery fish over time or among locations. Third, hatchery fish often have characteristics that reduce the effectiveness of stocking programs (Frankham 2008, Neff et al. 2011), but fitness has not been compared between wild and hatchery walleye in Black Bay. For example, hatchery fish might have low fitness if they came from an exogenous source that lacks important local adaptations (for examples of local adaptations in walleye see Galarowicz and Wahl [2003], Zhao et al. [2008]) or if they have become adapted to the hatchery environment (Sundström and Johnsson 2001, Araki et al. 2007). Resolving these uncertainties would provide a much better understanding of the outcome of the hatchery stocking program in Black Bay.

In this study we used genetic data to evaluate the contributions of wild sources and recent hatchery stocking events to the walleye population in Black Bay. Walleye of unknown ancestry were collected in Black Bay between 2007 and 2010, and in the lower Black Sturgeon River in 2007 and 2008. Population assignment and genetic clustering were used to assign these individuals to reference populations that included the wild populations in the Black Sturgeon River (above and below the dam), and the hatchery source populations from Cloud Lake and the St. Marys River. A previous study suggested that the upper and lower Black Sturgeon populations were genetically indistinguishable, but was based on fewer microsatellite loci (Wilson et al. 2007). To assess this further, we reexamined the relationship between these populations and the historical Black Bay population using nine microsatellite loci and additional samples from each location. To evaluate the contributions of the individual hatchery releases, we used genetic ancestry and age data to determine the abundance of hatchery fish in each of the 2003, 2004, and 2005 age classes. As walleye first mature at approximately age 3 in this population (Addison and Bobrowicz 2009), we used the frequency of hatchery ancestry in fish at least three years younger than the hatchery cohorts to estimate natural recruitment by hatchery fish. We also compared size and habitat associations in walleye from different sources to explore phenotypic differences that might influence recruitment by wild and hatchery fish. Together these data will provide a detailed profile of the current state of the walleye population in Black Bay that can be used to guide future management programs.

METHODS

Sample collection

All tissue samples were provided by the Ontario Ministry of Natural Resources (OMNR) Upper Great Lakes Management Unit (Thunder Bay, Ontario, Canada), and consisted of either dried scales or fin clips preserved in ethanol. Samples from the reference populations were collected in 2003 and 2004 at four locations (Fig. 1): the upper Black Sturgeon River ($n =$

89), the lower Black Sturgeon River ($n = 70$), Cloud Lake ($n = 147$), and the St. Marys River ($n = 56$). Samples from Cloud Lake and the St. Marys River were collected from adult fish spawned at the hatcheries, whereas samples from the Black Sturgeon River were collected as previously described by Wilson et al. (2007).

Walleye of unknown ancestry were collected in Black Bay using several techniques. The historical population was examined with archived scale samples from the 1966 Black Bay commercial walleye fishery ($n = 51$; Wilson et al. 2007). In 2007, walleye were collected in Black Bay as part of a test fishery for yellow perch (*Perca flavescens*) ($n = 116$; Addison 2008). In 2008, walleye were collected in Black Bay using the standardized fall walleye index netting (FWIN) protocol ($n = 191$; Addison and Bobrowicz 2009). In 2009, samples were obtained from walleye caught as bycatch of commercial fisheries in Black Bay ($n = 18$). In 2010, walleye were again collected in Black Bay using the FWIN protocol ($n = 99$). Additionally, walleye were collected from the lower Black Sturgeon River in 2007 using trapnets ($n = 83$) and in 2008 using a standardized riverine index netting (RIN) protocol ($n = 116$; Bobrowicz and Addison 2008).

Total length and body mass were measured for the subset of walleye sampled in the 2007 perch fishery, 2008 FWIN, 2008 RIN, and 2010 FWIN. Condition factor of these fish was calculated as $(\text{body mass}/\text{total length}^3) \times 10^5$ (Ricker 1975). An otolith, dorsal spine or scale sample was also collected from these fish for age determination by the OMNR fish ageing laboratory (Dryden, Ontario, Canada).

Genetic analysis

DNA was extracted from all samples using a simple lysis method (Wilson et al. 2007). Ten microsatellite loci were amplified for each fish using previously described primers comprising *Svi4*, *Svi6* (Borer et al. 1999); *Svi2*, *Svi7*, *Svi14* (Eldridge et al. 2002); *SviL4*, *SviL5*, *SviL6*, *SviL7*, and *SviL8* (Wirth et al. 1999). Microsatellite products were visualized using an AB 3730 sequencer (Applied Biosystems, Carlsbad, California, USA), and allele sizes determined by comparison to a ROX 350 size standard (Applied Biosystems). Alleles were scored using Genotyper 2.5 software (Applied Biosystems) and confirmed by manual inspection of the chromatograms. The presence of null alleles was assessed within each of the four reference populations using MicroChecker version 2.2.3 (Van Oosterhout et al. 2004). *SviL4* was the only locus that showed a significant heterozygote deficiency in all populations, consistent with the presence of a null allele; as a result, this locus was excluded from the subsequent genetic analyses.

For each population, heterozygosity measures were calculated using GenAlEx 6 (Peakall and Smouse 2006) and the average number of alleles and allelic richness adjusted to a common sample size of 30 genes were calculated using HP-Rare 1.0 (Kalinowski 2005). Pair-wise genetic divergence among the reference populations

and the historical Black Bay population were estimated using Φ_{st} (Excoffier et al. 1992) and Nei's D genetic distance (Nei 1972), both calculated using GenAlEx 6 (Peakall and Smouse 2006). Due to inconsistent amplification of some loci in the historical samples, allelic richness, heterozygosity, and pairwise divergence measures were calculated using only the six loci for which data were at least 50% complete in each population.

Individuals from all locations were grouped into genetic clusters using the Bayesian clustering method implemented in the program Structure version 2.3.3 (Pritchard et al. 2000). This analysis focused on numbers of genetic clusters (K) ranging from 1 to 6, with 5 replicate simulations run at each value of K to check for result convergence on a single most likely solution. All simulations were run for an initial burn-in of 100 000 steps followed by 200 000 resampling iterations. To maximize conservative testing and avoid positive bias, all simulations were run without a priori information on sampling locations and used model conditions that allowed for potential admixture and assumed independence among loci (uncorrelated allele frequencies). For each individual, the cluster with the highest membership coefficient was defined as the within-population membership coefficient (q). Low within-population membership coefficients may indicate fish that have parents from two different source populations, or fish that have a genotype that is relatively common in more than one potential source population, which causes them to show probabilistic memberships in multiple groups. For population pairs where interbreeding was geographically possible, we identified individuals as potential hybrids if their membership coefficients for both populations were at least 0.25, and the sum of the two membership coefficients was at least 0.8. The latter criterion was included to minimize the number of individuals that were identified as hybrids because they showed probabilistic membership in multiple groups. Potential mixed ancestry was explicitly considered for age classes at least three years after the first stocking events to test for potential reproductive contributions of stocked fish based on the earliest probable date of maturity of stocked males and females (Bozek et al. 2011).

Walleye of unknown ancestry were subsequently assigned to potential source populations using a combination of the membership coefficients from the Structure analysis and the Bayesian method of Rannala and Mountain (1997) as implemented in the software GeneClass2 (Piry et al. 2004). An individual was assigned to a reference population if the corresponding Structure membership coefficient (q) was greater than 0.80, or if the corresponding score in GeneClass2 was greater than 90. Individuals that failed to meet either assignment criteria (153 of 1036 individuals) or for which the Structure and GeneClass2 assignments disagreed (6 of 1036 individuals), were considered to have an unresolved origin. The accuracy of the

population assignments were evaluated using a self-assignment test in GeneClass2, in which each individual from the reference populations was assigned to a population as though its origin was unknown. The accuracy of the population assignments using the Structure membership coefficients and a combination of the two criteria were likewise evaluated using self-assignment of samples from the reference populations.

Statistical analysis

We used the 2008 data from Black Bay and the lower Black Sturgeon River to compare the habitat associations of walleye with different ancestries. For each habitat (bay, river), we summed the number of 3- and 4-yr-old fish from each major source (lower Black Sturgeon, upper Black Sturgeon, and St. Marys River). Fisher's exact tests were then used to evaluate differences in habitat association between each pair of sources (multiple-comparison adjusted $\alpha = 0.017$).

Restricted maximum-likelihood (REML) models were used to examine the factors that affected body mass, total length, and condition. Each model included habitat (bay, river), ancestry (lower Black Sturgeon, upper Black Sturgeon, St. Marys River), habitat $\times$ ancestry, and sampling year (random) as factors. This analysis included only 3-yr-old fish, as they were the only age class that was well represented in all combinations of ancestry and habitat. However, similar patterns of significance were observed when age was included as a covariate and all individuals were analyzed instead. Statistical analyses were performed using JMP version 4.0.4 (SAS Institute 2000).

RESULTS

All microsatellite loci were highly polymorphic, with an average of at least six alleles per locus observed in all populations (Table 1). Adjusted for sample size, allelic richness was highest in the St. Marys River, and lowest in the lower Black Sturgeon River (Table 1). Allelic richness was higher in the historical Black Bay population than the current populations in both the upper and lower Black Sturgeon River (Table 1). In all populations the observed microsatellite heterozygosity was lower than the expected heterozygosity (Table 1).

Significant genetic differentiation was observed between all reference populations (Table 2). The smallest genetic distance was observed between the upper and lower Black Sturgeon River populations, whereas the largest genetic distance was observed between the St. Marys River and Cloud Lake (Table 2). Pairwise Φ_{st} comparisons (Excoffier et al. 1992) revealed similar relationships among populations (Table 2). The historical Black Bay population was genetically distinct from all of the reference populations (Table 2), although the apparent divergences may have been amplified by incomplete genotypes (missing data) from the historical samples that altered allele frequencies.

TABLE 1. Summary of genetic results for walleye (*Sander vitreus*), by sampling location and year.

Sampling location	Year	<i>N</i>	<i>A</i>	<i>A_R</i>	<i>H_E</i>	<i>H_O</i>	<i>q</i>
Lower Black Sturgeon River	2004	70	6.50	4.96	0.657	0.616	0.848
Upper Black Sturgeon River	2004	89	7.17	5.38	0.694	0.673	0.880
St. Marys River	2004	56	9.67	7.20	0.743	0.676	0.790
Cloud Lake	2003	147	8.83	6.04	0.737	0.687	0.880
Black Sturgeon River	2007	83	7.50	5.47	0.668	0.614	0.821
	2008	116	8.33	5.34	0.676	0.655	0.868
Black Bay	1966	51	7.67	6.19	0.676	0.622	0.732
	2007	116	10.33	6.77	0.743	0.684	0.832
	2008	191	9.67	5.94	0.716	0.648	0.864
	2009	18	6.17	5.87	0.629	0.546	0.857
	2010	99	9.33	5.97	0.699	0.606	0.880

Notes: Data comprise sample size (*N*), average number of alleles (*A*), allelic richness adjusted to a common sample size (*A_R*), expected heterozygosity (*H_E*), observed heterozygosity (*H_O*), and mean within-population membership coefficient (*q*).

The Structure simulations reached a plateau in likelihood values when *K* (number of genetic clusters) = 4 [$\ln P(K|D) = -29.197$], which indicated the presence of four major genetic clusters, with each cluster corresponding to one of the reference populations (Fig. 2). This clustering pattern was robust to changes in the simulation parameters, which included assuming correlated allele frequencies or incorporating prior information on sampling locations (data not shown). The membership coefficients (*q*) indicated that 57% of the historical samples were more closely related to the Cloud Lake population than the other populations (Fig. 2), which suggests that the historical Black Bay population may have been genetically similar to the current population in Cloud Lake. However, 43% of the historical samples were more closely related to other populations (Fig. 2), and the historical samples had low within-group membership coefficients (Table 2), so few definitive conclusions can be drawn from the historical data. The recent walleye samples from Black Bay and the Black Sturgeon River typically had within-population membership coefficients that were consistent with ancestry in a single cluster (Table 1), which suggests that there has been minimal interbreeding between fish from different sources in this area. However, 26 individuals in Black Bay and the lower Black Sturgeon River showed potential mixed ancestry from walleye above and below the barrier, indicating that fish that migrated downstream over the dam may have interbred with the below-

barrier population (Fig. 2). Most walleye from Black Bay and the Black Sturgeon River were members of the lower Black Sturgeon River cluster, followed by the St. Marys River and upper Black Sturgeon River clusters (Fig. 2). Cloud Lake ancestry was rarely inferred for these fish (Fig. 2). The current walleye population in Black Bay and the Black Sturgeon River thus consists mainly of wild fish from the Black Sturgeon River and hatchery fish from the St. Marys River.

The substantial genetic differences among the potential source populations enabled confident inference of individual ancestry. A self-assignment test using GeneClass2 resolved the ancestry for 80.4% of the samples (assignment scores above 90), with 96.9% of those assignments being correct. Self-assignment using Structure resolved the ancestry for 76.0% of the samples (*q* values above 0.8), with 96.4% of those assignments being correct. Combining these two assignment criteria resulted in more samples with resolved ancestry (85.4%), and 96.1% correct assignment (i.e., assigned to their original sampling locale) for fish that assigned to a single source (Table 3). Consequently, the combined criteria were used for all ancestry assignments. The self-assignment tests all likely underestimated assignment accuracy, as the majority of the incorrect assignments were upper Black Sturgeon River individuals identified in the lower Black Sturgeon River (Table 3), where they potentially represent legitimate migrants that passed over the Black Sturgeon Dam (Fig. 1). The unresolved

TABLE 2. Pairwise genetic divergences among the set of potential source walleye populations, with Φ_{st} values (Excoffier et al. 1992) above the diagonal and genetic distance (Nei's [1972] *D*) below the diagonal.

Population	Lower Black Sturgeon River	Upper Black Sturgeon River	St. Marys River	Cloud Lake	Historical Black Bay
Lower Black Sturgeon River		0.083	0.098	0.125	0.156
Upper Black Sturgeon River	0.119		0.148	0.114	0.162
St. Marys River	0.164	0.293		0.139	0.168
Cloud Lake	0.207	0.203	0.313		0.120
Historical Black Bay	0.253	0.283	0.358	0.223	

Notes: Values are based on the six microsatellite loci for which genotypes were determined for at least 50% of the individuals in the historical Black Bay population. All pairwise Φ_{st} values were significant at $P < 0.001$ based on 999 randomizations.

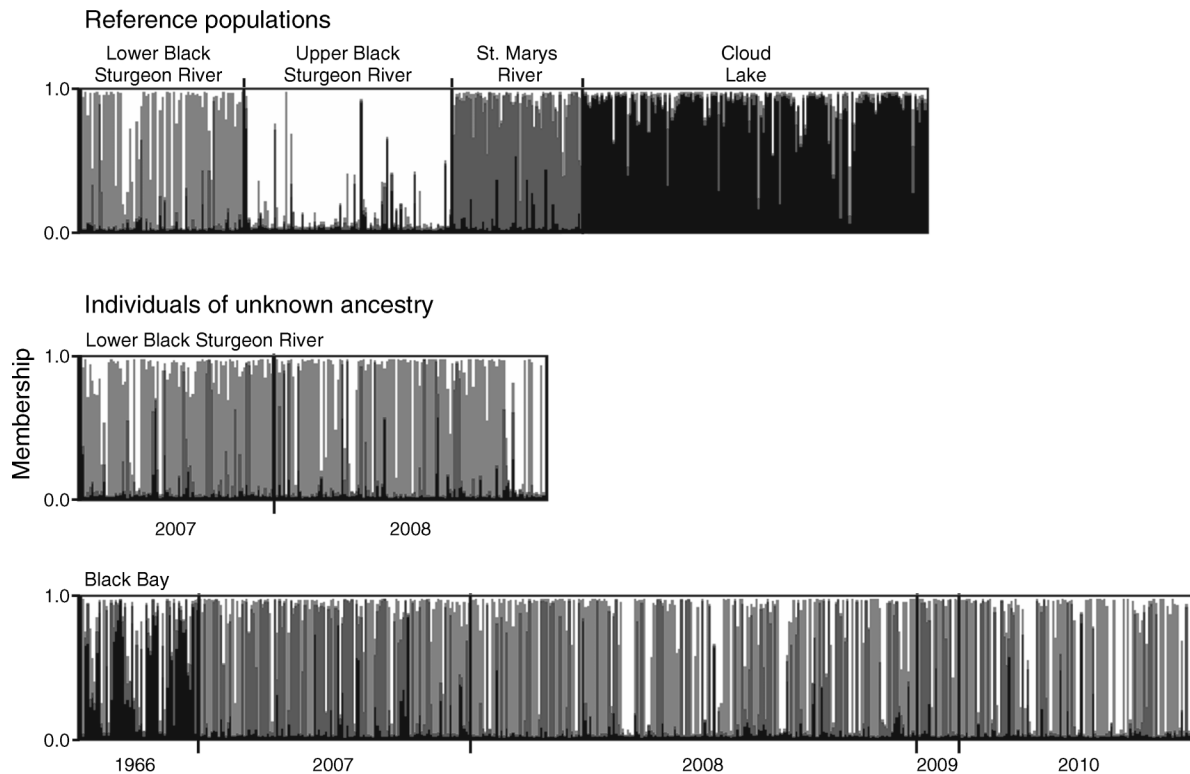


FIG. 2. Results of the Structure analysis (version 2.3.3; Pritchard et al. 2000) of walleye samples from Lake Superior showing the optimal solution with four genetic clusters. Each vertical line represents one individual, with different shades of gray indicating proportional membership in each of the four genetic clusters identified by Structure.

ancestry for walleye that could not be assigned to a single source based on the combined criteria could result from incomplete genotypes, possessing common (shared) alleles at some loci, or mixed ancestry. Of these, the last possibility is unlikely within the different source populations other than the lower Black Sturgeon River, due to their reciprocal isolation and geographic separation (Fig. 1).

Genetic analysis of individual ancestry for walleye from Black Bay and the Black Sturgeon River showed that multiple sources contributed to the current population in this area (Table 4). More than half of the walleye were assigned to the lower Black Sturgeon River population, which made this population the largest

source of walleye in both Black Bay and the Black Sturgeon River (Table 4). The upper Black Sturgeon River population provided a second major source of wild-origin fish, with approximately one sixth of the walleye sampled in Black Bay and the Black Sturgeon River originating from this population. Less than 2% of the walleye sampled in Black Bay and the Black Sturgeon River were assigned to the Cloud Lake population, indicating that the 2003 stocking event had little lasting impact. In contrast, approximately one quarter of the sampled walleye were assigned to the St. Marys River population, indicating that the 2004 and 2005 stocking events contributed substantially to the walleye population in this area. The proportion of fish

TABLE 3. Accuracy of self-assignment for walleye samples from the reference populations.

True population origin of samples	Population assignment				
	Lower Black Sturgeon River	Upper Black Sturgeon River	St. Marys River	Cloud Lake	Unresolved†
Lower Black Sturgeon River	52	7	1	0	10
Upper Black Sturgeon River	1	76	0	1	11
St. Marys River	0	0	42	1	13
Cloud Lake	1	0	0	127	19

† “Unresolved” indicates individuals that were not assigned to any single population with a GeneClass2 score above 90 or a Structure membership coefficient above 0.80.

TABLE 4. Number of walleye assigned to each reference population using microsatellite data.

Sampling location	Year	Source populations to which individual walleye assigned				
		Lower Black Sturgeon River	Upper Black Sturgeon River	St. Marys River	Cloud Lake	Unresolved†
Black Sturgeon River	2007	53	6	7	1	16
	2008	73	20	7	1	15
Black Bay	2007	41	7	46	3	19
	2008	88	32	44	1	26
	2009	8	0	9	0	1
	2010	47	24	17	2	9

† “Unresolved” indicates individuals that were not assigned to any single population with a GeneClass2 score above 90 or a Structure membership coefficient above 0.80.

with unresolved ancestry was slightly lower in Black Bay and the Black Sturgeon River (13.8%) than in the reference populations (14.6%), which is consistent with minimal interbreeding between source populations in this area.

Incorporating age information provided even greater resolution of the contributions of the different hatchery stocking events (Fig. 3). All of the walleye that had Cloud Lake ancestry came from age classes that were inconsistent with the 2003 hatchery release, thus

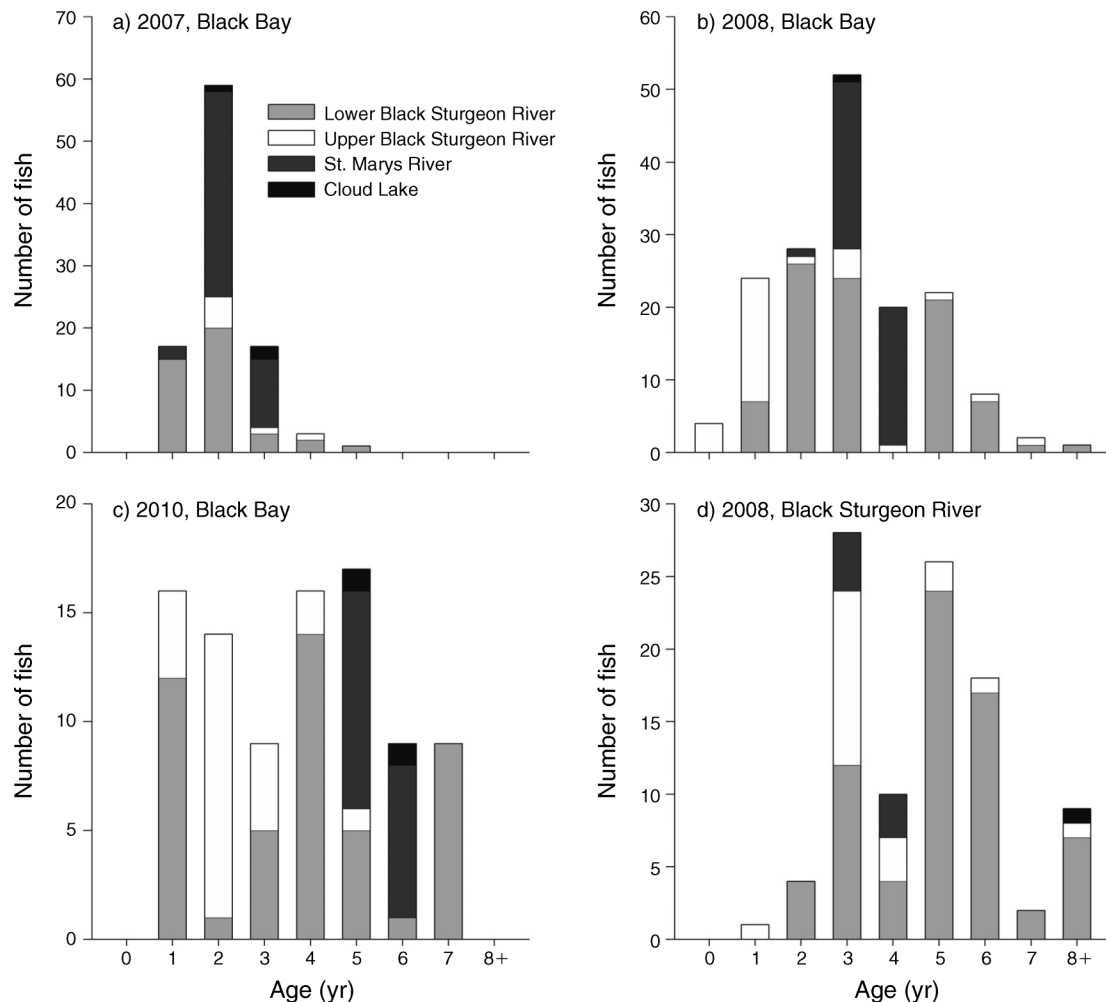


FIG. 3. Ancestry of walleye from Black Bay and the lower Black Sturgeon River. The grayscale bars show the number of walleye of each age that were assigned to each reference population (individuals of unresolved ancestry not shown). The panels portray the various sampling events and show walleye collected from (a) the 2007 perch fishery in Black Bay, (b) the 2008 fall walleye index netting (FWIN) in Black Bay, (c) the 2010 FWIN in Black Bay, and (d) the 2008 riverine index netting (RIN) in the lower Black Sturgeon River.

TABLE 5. Summary of models comparing the effects of habitat (bay, river) and ancestry (lower Black Sturgeon River, upper Black Sturgeon River, St. Marys River) on the size of 3-yr-old walleye.

Trait	ANOVA term	df	F	P†
Body mass	Habitat	1, 93	71.54	<0.001
	Ancestry	2, 93	13.77	<0.001
	Ancestry × Habitat	2, 93	3.99	0.022
	Year (random)	2, 93	4.23	0.017
Total length	Habitat	1, 95	30.45	<0.001
	Ancestry	2, 95	14.37	<0.001
	Ancestry × Habitat	2, 95	5.04	0.008
	Year (random)	2, 95	1.30	0.28
Condition	Habitat	1, 93	61.72	<0.001
	Ancestry	2, 93	4.86	0.010
	Ancestry × Habitat	2, 93	3.69	0.029
	Year (random)	2, 93	0.15	0.86

† P values shown in bold indicate significance for $\alpha = 0.05$.

providing no evidence that this stocking event contributed to the walleye population in Black Bay. In contrast, both the 2004 and 2005 hatchery releases of St. Marys River fish contributed substantially to their respective age classes (Fig. 3). Across years and sampling locations, 71% of the 2004 age-class and 45% of the 2005 age class originated from the St. Marys River hatchery releases. However, neither full nor mixed St. Marys River ancestry was inferred for any of the 68 individuals from the 2007–2009 age classes, suggesting that hatchery fish did not contribute substantially to natural recruitment within these cohorts.

Analysis of the combined 2004 and 2005 age classes showed that walleye with different origins differed significantly in their habitat associations. Fish from the upper Black Sturgeon River were observed less often in the bay than in the river (river, 15; bay, 5), whereas walleye from the lower Black Sturgeon River were observed more often in the bay than the river (river, 16; bay, 24), and walleye from the St. Marys River were observed in much higher numbers in the bay than the river (river, 7; bay, 42) (pairwise Fisher's exact tests, all $P < 0.015$).

All size characteristics (body mass, total length, and condition) showed consistent effects of habitat and ancestry in 3-yr-old walleye (Table 5; Fig. 4). Fish sampled in Black Bay were significantly heavier, longer, and in better condition than fish sampled in the lower Black Sturgeon River (Table 5; Fig. 4). Additionally, fish from the lower Black Sturgeon River and St. Marys River populations were significantly heavier, longer, and in better condition than fish from the upper Black Sturgeon River population (Table 5; Fig. 4). We also observed a significant interaction between ancestry and habitat, in which the size differences between upper Black Sturgeon River fish and the other populations were smaller when all fish were sampled in Black Bay than when all fish were sampled in the Black Sturgeon River. Lastly, there was a significant effect of sampling

year on body mass (Table 5), indicating significant annual variation in walleye size.

DISCUSSION

The genetic evidence showed that walleye in Black Bay were primarily of wild origin, with most descended from the upper or lower Black Sturgeon River

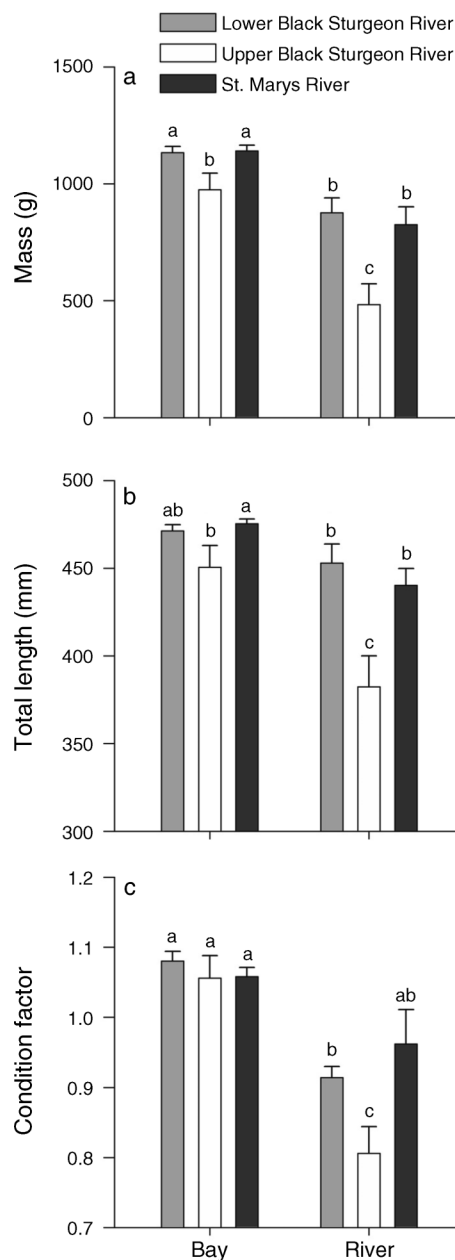


FIG. 4. Size and condition of 3-yr-old walleye from Black Bay and the lower Black Sturgeon River. Data (means ± SE) are presented for (a) mass, (b) total length, and (c) condition. The condition factor of these fish was calculated as (body mass/total length³) × 10⁵ (Ricker 1975). The gray shade of the bars indicates the ancestry inferred for each individual using genetic data. Bars with different lowercase letters above are significantly different among groups (at $\alpha = 0.05$).

populations. The populations in the upper and lower Black Sturgeon River were more closely related to each other than to the other reference populations, consistent with recent divergence of these populations following the construction of the Black Sturgeon River (Camp 43) dam. The recent hatchery stocking events differed considerably in their effectiveness: the 2003 release of fry from Cloud Lake made no contribution to the walleye population in Black Bay, whereas the 2004 and 2005 releases of fingerlings from the St. Marys River each made a substantial contribution to their age classes. Hatchery fish were similar in size to wild fish, but rarely utilized habitat in the lower Black Sturgeon River and appeared to have lower reproductive success than wild fish.

Despite historical overexploitation and habitat fragmentation, natural populations of walleye have persisted in the Black Bay area. Walleye above and below the Camp 43 dam appear to have diverged through reduced population sizes and genetic drift in the 50 years since the dam's construction. Studies of other fishes have shown that dams represent important migration barriers that can shape patterns of genetic differentiation over relatively short time scales (e.g., Meldgaard et al. 2003, Yamamoto et al. 2004, Clemento et al. 2009). Over the past half century, both Black Sturgeon River populations would have had reduced population sizes, as walleye above the dam were limited by foraging habitat, and those below the dam were limited by severely reduced spawning habitat and the commercial fishery. Indeed, both populations in the Black Sturgeon River had lower genetic diversity than the historical Black Bay population, consistent with population bottlenecks or reduction (Nei et al. 1975, England et al. 2003). Genetic drift can thus explain the differentiation observed between the upper and lower Black Sturgeon River walleye populations, with their relative similarity compared to the other reference populations consistent with their recent differentiation.

The results of our genetic analyses differed from those of a previous study of walleye in Black Bay (Wilson et al. 2007). We observed significant genetic differentiation between walleye in the upper and lower Black Sturgeon River, and an unresolved relationship between the historical Black Bay samples and the other populations. In contrast, Wilson et al. (2007) observed no genetic differentiation between walleye in the upper and lower Black Sturgeon River or between historical and contemporary Black Bay samples. The greater genetic structure observed in our study likely resulted from the greater power of our analyses, which examined 210 samples at nine microsatellite loci, compared to 111 samples at four microsatellite loci in the previous study. Low DNA integrity of the historical samples led to incomplete microsatellite genotypes in both studies, which likely contributed to the uncertainty in resolving the relationship between the historical Black Bay samples and the contemporary populations. We did observe a weak but

consistent association between the historical Black Bay population and the Cloud Lake population, which suggests that walleye in Black Bay were historically more similar to walleye in nearby Thunder Bay (the source of walleye introduced into Cloud Lake). The previous study would not have observed this relationship because it did not include samples from the Thunder Bay population (Wilson et al. 2007). Overall, these differences emphasize the importance of sampling design when evaluating genetic structure.

Walleye restoration programs often benefit from mitigating the effects of dams on habitat connectivity (e.g., Gillenwater et al. 2006, MacDougall et al. 2007). Walleye typically return to natal sites to spawn (Jennings et al. 1996, Stepien and Faber 1998, Palmer et al. 2005), so dams that provide an impassible barrier to migration can prevent walleye from reaching spawning sites. Walleye currently congregate below the Black Sturgeon dam during the spring spawning migration (Furlong et al. 2006, Bobrowicz 2010), which suggests that the dam is a barrier to upstream migration that limits access to spawning habitat. A lack of spawning habitat may in turn explain why the Black Bay walleye stock has not recovered despite long-term closure of the fishery. Interestingly, substantial numbers of walleye from the upper Black Sturgeon River population are present below the dam, which indicates that the dam is only a weak barrier to downstream dispersal. Dispersal from the upper Black Sturgeon River population thus has the potential to increase the abundance of walleye in Black Bay, although this effect is likely to be limited by the small size of the upstream population. Historically, walleye could make use of the habitat continuum in Black Bay and its major tributary (forage base in Black Bay, adult spawning and juvenile nursery habitat in the Black Sturgeon River), and the productive capacity of the intact system appears to have been much greater than the sum of its fragmented parts. Removing the Black Sturgeon River dam would thus restore habitat connectivity and contribute to the recovery of the natural walleye population in Black Bay.

Individual hatchery stocking events had highly variable contributions to the walleye population in Black Bay. The 2003 release of Cloud Lake fry did not contribute any observable number of fish. Although Cloud Lake ancestry was inferred in a small number of fish, none of those fish had ages that were consistent with the 2003 hatchery release. Instead, fish in Black Bay that had Cloud Lake ancestry were likely natural migrants from nearby Thunder Bay (itself the source of walleye in Cloud Lake). In contrast, the 2004 and 2005 releases of St. Marys River fingerlings both contributed substantially to their age classes. It is possible that the release of Cloud Lake fish was unsuccessful because this population was not adapted to the local conditions in Black Bay, whereas fish from the Black Sturgeon River and St. Marys River were adapted to these conditions (for examples of local



PLATE 1. Aerial photograph of the Black Sturgeon dam, taken in 2003. Photo credit: Mike Friday, Ontario Ministry of Natural Resources.

adaptations in walleye see Galarowicz and Wahl [2003], Zhao et al. [2008]). However, successful migrants with Cloud Lake ancestry were observed in Black Bay, so the absence of local adaptations is unlikely to explain the complete failure of the 2003 hatchery release. Instead, the differing success of the hatchery releases likely resulted from the age of the hatchery fish: in 2003 walleye were released as fry (post-hatch, prior to feeding), and in 2004 and 2005 walleye were released as summer fingerlings (more than a month older, fed at the hatchery). Multiple studies have found that walleye stocked as fingerlings have much higher survival than walleye stocked as fry (e.g., Fielder 1992, Paragamian and Kingery 1992), which has led to a recent increase in the proportion of walleye stocked at the fingerling stage (Kerr 2011). The differences that we observed among hatchery releases were consistent with low survival of hatchery fry, and showed that more consistent contributions were achieved when walleye were stocked as fingerlings.

Reduced fitness is a common concern in hatchery fishes, as it can limit the effectiveness of stocking programs (Frankham 2008, Neff et al. 2011). In our present study we used a variety of measures to evaluate the fitness of hatchery fish in Black Bay. We found that wild and hatchery walleye were similar in size, which suggests that these fish had similar growth rates and ability to feed on natural prey items. Adult survival was also comparable between wild and hatchery fish, as the

relative abundance of hatchery fish within age classes was largely unchanged between 2007 and 2010. Hatchery ancestry was not inferred for any of the walleye that were at least 3 years younger than the hatchery cohorts, thus providing no evidence for natural reproduction by hatchery fish. It is still too early to draw firm conclusions about the reproductive fate of hatchery fish in Black Bay, as stocked fish may have taken longer to mature based on local growing degree days (Bozek et al. 2011), in which case their offspring might not yet be detectable using the netting methods that were employed. Hatchery fish were rarely present in the Black Sturgeon River, the most commonly used spawning habitat available in this area (Bobrowicz 2010), suggesting a potential reason for low reproductive success by stocked fish. Ironically, as the native walleye population has persisted in Black Bay, low reproductive contributions of stocked fish may be beneficial by limiting the spread of exogenous genes into the native population.

Fish that utilize lake habitats are typically larger than fish that utilize river habitats, even when those fish are members of the same species (Hutchings 1986, Randall et al. 1995). These differences are thought to arise because lake habitats reduce the energy required to maintain position, offer more abundant food supplies, and provide greater access to large, energetically profitable prey items (Hutchings 1986, Keeley and Grant 2001). Walleye in Black Bay were larger than walleye in the lower Black Sturgeon River, which is

consistent with lake habitat being more energetically profitable than river habitat. Walleye from the upper Black Sturgeon River population were smaller than walleye from the other populations, which likely reflects limited access to lake habitat prior to their migration over the Black Sturgeon River dam. Overall, we found that body size in walleye was more closely associated with habitat than with ancestry.

The current status of walleye in Black Bay is the product of a history of anthropogenic influences. Overexploitation by the commercial fishery was associated with a massive decline in abundance more than 40 years ago, with signs of improvement occurring only within the last 5 years. Construction of the Black Sturgeon River Dam contributed to the collapse of the historical population (Furlong et al. 2006), and has had a major effect on the genetic structure of the remaining population fragments by limiting population sizes and restricting gene flow in a previously connected system. Even the restoration programs have left a signature of anthropogenic involvement, with large numbers of genetically divergent hatchery fish currently residing in Black Bay. Improvements in the abundance of walleye have occurred independently of hatchery stocking, with 75% of the current walleye population in Black Bay having a wild origin. Consequently, any further exogenous stocking is almost certainly a counterproductive use of management resources. Instead, the removal of an ageing dam that artificially restricts migration offers the greatest potential to restore formerly productive walleye habitat (Gillenwater et al. 2006, Bobrowicz et al. 2010).

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